

TOWARDS LIOUVILLE QUANTUM GRAVITY ON FRACTALS

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ABSTRACT. Motivated by the recent construction of fractional Gaussian fields on the Sierpinski gasket, we study those Gaussian fields whose covariance function exhibits a logarithmic behavior. We introduce a parametric family of random measures associated with these fields, which can be regarded as the analogue to Liouville quantum gravity. The construction is based on Kahane's approach via Gaussian multiplicative chaos. We establish reflection invariance and scaling self-similarity for both the fields and the measures, and in addition construct the corresponding analogue to Liouville Brownian motion.

CONTENTS

1. Introduction	2
2. Background and notation	3
2.1. The Sierpinski gasket	3
2.2. Riesz potentials	4
2.3. Function spaces	6
3. Fractal Gaussian free fields	7
3.1. Approximating Gaussian fields	8
3.2. Invariance properties	10
4. Liouville Quantum Gravity measure	12
4.1. Approximations	13
4.2. Liouville Quantum Gravity measure	15
4.3. Invariance properties	17
5. Liouville Brownian motion	18
5.1. Capacities and smooth measures	19
5.2. LBM on the Sierpinski gasket	21
References	22

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1. INTRODUCTION

In mathematical physics, the Liouville quantum gravity (LQG) measure plays a central role in the study of two-dimensional quantum gravity and the Knizhnik-Polyakov-Zamolodchikov (KPZ) formula, see e.g. [17, 9, 5, 8]. From a mathematical viewpoint, the LQG measure (e.g. in $\mathbb{R}^2$) may formally be expressed as

$$e^{\gamma h(x)} dx,$$

where h is a Gaussian free field (GFF). This expression only holds formally since the field cannot be defined pointwise due to the logarithmic behavior of its covariance function $\mathbb{E}[h(x)h(y)]$.

One way to construct the LQG measure consists in applying the *Gaussian multiplicative chaos* (GMC) method, which goes back to Kahane [15]. There, a martingale approximation was employed, assuming that the covariance kernel $\mathbb{E}[h(x)h(y)]$ is decomposable into a sum of non-negative kernels. This condition is hard to check in practice, which lead to further investigations by Rhodes and Vargas in [20], where they approximated the fields via convolution with appropriate mollifiers. This approach yields a random measure that is unique in law, but not as a function of the underlying field. A related construction using circle averages was proposed by Duplantier and Sheffield in [9] for the special case of a Gaussian free field in two dimensions. An alternative approach based on thick points of the underlying field can be found in [4].

To a certain extent and from a theoretical viewpoint, one may speak of an LQG measure in any metric measure space, in particular beyond the Euclidean setting, as soon as there is a meaningful concept of log-correlated GFF. Constructions of such fields have been investigated for the torus in [6], in the Riemannian setting in [7], and also explored in the abstract framework of Dirichlet form theory in [11]. In the present paper, we rely on a characterization of GMC through a transformation due to Shamov [22], which had already been pointed out in [21]. The recent construction of GFFs in the context of fractals, and specifically on the Sierpinski gasket in [3, 2], has sparked the interest in investigating the associated LQG measure here.

From a more geometric viewpoint, the LQG measure can heuristically be viewed as the random volume measure associated with the formal random metric $e^{\gamma h(x)} |dx|^2$ of Liouville quantum gravity. Garban-Rhodes-Vargas introduced in [13] the *Liouville Brownian motion* (LBM) in $\mathbb{R}^2$ as the natural diffusion associated with this random geometry, which was independently considered by Berestycki in [4]. Subsequently, alternative constructions using Dirichlet forms have been proposed in [23, 11] and in [7] in the context of Riemannian manifolds. In the present paper, we follow the general approach via Dirichlet form theory to define this process as a time change of the Brownian motion on the Sierpinski gasket.

The paper is organized as follows: After introducing in Section 2 notations and basic geometric and analytic results concerning the Sierpinski gasket, Section 3 deals with the construction and (invariance) properties of the random field which we interpret as the analogue of the planar GFF. That field is used in Section 4 to construct the associated LQG measures by approximation, see Theorem 10. It is worth pointing out that these measures are indexed by a parameter $\gamma > 0$ that lies in a certain, non-optimal, regime, discussed in Remark 4. Finally, Section 5 is devoted to the construction of the LBM associated with a given LQG

measure on the gasket, see Theorem 19. The result relies on the key property of the LQG measure being $\mathbb{P}$ -a.s. smooth.

2. BACKGROUND AND NOTATION

2.1. The Sierpinski gasket. In this section we review notation and basic results concerning the Sierpinski gasket that are of relevance to the purpose of the paper. For further details we refer the reader e.g. to [24, Chapter 3].

The standard Sierpinski gasket K , see Figure 1, is the unique compact subset of $\mathbb{R}^2$ that satisfies the fixed-point equation

$$K = \bigcup_{i=1}^3 F_i(K),$$

where $F_i: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are the mappings

$$F_i(p) := \frac{1}{2}(p - p_i) + p_i$$

and $V_0 := \{p_1, p_2, p_3\}$ denotes the set of vertices of an equilateral triangle of side length one. The set V_0 is also regarded as the natural boundary of K . As a metric measure space, K is equipped with the Euclidean metric and the standard self-similar Bernoulli measure μ that gives the same weight to each triangular cell of the same side-length. This measure thus satisfies

$$\mu(F_w(V_0)) = \frac{1}{3^m}$$

for any $F_w := F_{w_1} \circ \dots \circ F_{w_m}$, where $w = w_1 \dots w_m \in \{1, 2, 3\}^m$, $m \geq 0$. The latter measure is comparable to the normalized d_H -dimensional Hausdorff measure, where $d_H = \frac{\log 3}{\log 2}$ is the (Euclidean) Hausdorff dimension of K . For simplicity, and since K is compact, we set both its diameter and measure to be 1.

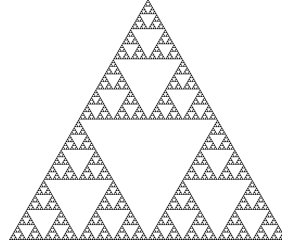


FIGURE 1. The standard Sierpinski gasket.

Regarding functions on K , we will denote by $L^2(K) := L^2(K, \mu)$ the space of square-integrable functions with respect to the measure μ , and by $\langle u, v \rangle$ the associated standard L^2 -inner product with norm $\|u\|^2 := \langle u, u \rangle$.

The Dirichlet Laplacian on K is the self-adjoint operator on $L^2(K)$ that is the infinitesimal generator of the local and regular Dirichlet form $(\mathcal{E}, \mathcal{F})$ given by

$$\begin{cases} \mathcal{E}(u, v) := \lim_{n \rightarrow \infty} \left(\frac{5}{3}\right)^n \sum_{x, y \in V_n, x \sim y} (u(x) - u(y))(v(x) - v(y)), \\ \mathcal{F} := \{u \in C(K) : \mathcal{E}(u, u) < \infty\}. \end{cases} \quad (2.1)$$

Its associated heat semigroup $\{P_t\}_{t \geq 0}$ has a heat kernel $p_t(x, y)$ that satisfies the sub-Gaussian estimates

$$p_t(x, y) \asymp c_1 t^{-d_S/2} \exp \left(-c_2 \left(\frac{d(x, y)^{d_W}}{t} \right)^{\frac{1}{d_W-1}} \right) \quad (\text{HKE})$$

for μ -a.e. $x, y \in K$ and $t \in (0, 1)$. Here, d_S and d_W denote respectively the so-called spectral and walk dimension, which in the case of the Sierpinski gasket are

$$d_W = \frac{\log 5}{\log 2} \quad \text{and} \quad d_S = \frac{\log 9}{\log 5} = \frac{2d_H}{d_W},$$

see e.g. [18]. In addition, it is also known that the heat kernel satisfies the global estimate

$$p_t(x, y) \leq c_3 t^{-d_S/2} \exp(-t/2) \quad (2.2)$$

for μ -a.e. $x, y \in K$ and $t > 1$. We refer the reader to [18] and [1, Section 5] for further details.

The infinitesimal generator Δ associated with $\{P_t\}_{t \geq 0}$ is a non-positive operator with pure point spectrum that is regarded as the standard intrinsic Laplacian on K . The eigenvalues of $-\Delta$ can be described by means of a genealogy tree, where each distinct eigenvalue has a unique “genealogical line”.

We next mention some relevant features of the spectrum that will be used later on and refer the interested reader e.g. to [24, Section 3.3] for further details. Below, $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$ denote the ordered eigenvalues including multiplicity, and $\lambda_1^{(6)}$ the sixth lowest Dirichlet eigenvalue.

Proposition 1. *For any $j \geq 1$, define the numbers*

$$N_j := \frac{1}{2}(3^{j+1} - 3) \quad \text{and} \quad \Lambda_j := 5^{j-2} \lambda_1^{(6)}. \quad (2.3)$$

Then, for any $j \geq 2$,

- (i) $\lambda_{N_j} = \Lambda_j$, i.e. $\text{mult } \Lambda_j = N_j$.
- (ii) $\#\{k \geq 1: \varphi_k \text{ eigenfunction of } -\Delta \text{ with eigenvalue } \lambda_k \leq \Lambda_j\} = N_{j+1}$.

2.2. Riesz potentials. By virtue of the spectral theorem, the heat semigroup can be expressed as

$$P_t u(x) = \sum_{k=1}^{\infty} e^{-\lambda_k t} \langle u, \varphi_k \rangle \varphi_k(x).$$

The Riesz potential admits an equivalent expression in the L^2 -sense in terms of the spectral resolution of the heat semigroup, that is

$$(-\Delta)^{-s/2} u = \int_0^{\infty} \lambda^{-s/2} dE_{\lambda}(u) = \sum_{k=1}^{\infty} \lambda_k^{-s/2} \langle u, \varphi_k \rangle \varphi_k = \frac{1}{\Gamma(s/2)} \int_0^{\infty} t^{s/2} P_t u \frac{dt}{t}, \quad (2.4)$$

c.f. [14, Proposition 2.4]. The above can be regarded as an integral operator by writing in (2.4) the heat semigroup in terms of the heat kernel. In this way, we arrive at the definition of the Riesz kernel of order $s > 0$.

Definition 1. Let $s > 0$. The fractional Riesz kernel of order s is given by

$$k_s(x, y) := \frac{1}{\Gamma(s/2)} \int_0^\infty t^{s/2} p_t(x, y) \frac{dt}{t} \quad (2.5)$$

for all $x, y \in K$, $x \neq y$.

Remark 1. Note that, a priori, $k_s(x, y)$ need not be well-defined for all $s > 0$. When $0 < s < d_S$, it is known that the Riesz kernel (2.5) satisfies

$$C^{-1} d(x, y)^{-\frac{d_W}{2}(d_S-s)} \leq k_s(x, y) \leq C d(x, y)^{-\frac{d_W}{2}(d_S-s)},$$

see [14, Proposition 2.7, Proposition 2.8].

At criticality, i.e. when $s = d_S$, the kernel displays a similar logarithmic behavior as in the Euclidean/Riemannian manifold case when s equals the dimension of the space. The proof of this estimate can be found in [2, Proposition 2.8], and we include it here in detail for completeness.

Proposition 2. [2, Proposition 2.8] *There exists constants $C_0, c_0 > 0$ such that*

$$c_0 \log \frac{1}{d(x, y)} \leq k_{d_S}(x, y) \leq C_0 \left(1 + \log \frac{1}{d(x, y)} \right). \quad (2.6)$$

Proof. Setting $s = d_S$ in (2.5) we first separate

$$k_{d_S}(x, y) = \frac{1}{\Gamma(d_S/2)} \int_0^1 t^{d_S/2} p_t(x, y) \frac{dt}{t} + \frac{1}{\Gamma(d_S/2)} \int_1^\infty t^{d_S/2} p_t(x, y) \frac{dt}{t} = I_1 + I_2.$$

Applying the heat kernel estimates (HKE) and the change of variables $v = \frac{d(x, y)}{t^{1/d_W}}$,

$$\begin{aligned} I_1 &\leq c_1 \int_0^1 \exp \left(-c_2 \left(\frac{d(x, y)}{t^{1/d_W}} \right)^{\frac{d_W}{d_W-1}} \right) \frac{dt}{t} \\ &= d_W c_1 \int_{d(x, y)}^\infty \exp \left(-c_2 v^{\frac{d_W}{d_W-1}} \right) \frac{dv}{v} \\ &= d_W c_1 \int_{d(x, y)}^1 \exp \left(-c_2 v^{\frac{d_W}{d_W-1}} \right) \frac{dv}{v} + d_W c_1 \int_1^\infty \exp \left(-c_2 v^{\frac{d_W}{d_W-1}} \right) \frac{dv}{v} \\ &=: I_{1,1} + I_{1,2}. \end{aligned} \quad (2.7)$$

The first integral above is bounded by

$$I_{1,1} \leq d_W c_1 \int_{d(x, y)}^1 \frac{dv}{v} = d_W c_1 \left[\log v \right]_{d(x, y)}^1 = -d_W c_1 \log d(x, y) = d_W c_1 \log \frac{1}{d(x, y)}$$

and the second by a constant. For the second term in (2.7) one applies the estimate (2.2) for long times to get

$$I_{1,2} \leq c_3 \int_1^\infty e^{-t/2} \frac{dt}{t} =: c_4 < \infty$$

In total we arrive at

$$k_{d_S}(x, y) \leq d_W c_1 \log \frac{1}{d(x, y)} + c_5.$$

For the lower bound, we apply the corresponding lower bound of the heat kernel (HKE) and argue as in (2.7) to get

$$\begin{aligned}
k_{d_S}(x, y) &\geq C_1 \int_0^1 \exp\left(-C_2 \left(\frac{d(x, y)}{t^{1/d_W}}\right)^{\frac{d_W}{d_W-1}}\right) \frac{dt}{t} \\
&= d_W C_1 \int_{d(x, y)}^\infty \exp\left(-C_2 v^{\frac{d_W}{d_W-1}}\right) \frac{dv}{v} \\
&\geq d_W C_1 \int_{d(x, y)}^1 \exp\left(-C_2 v^{\frac{d_W}{d_W-1}}\right) \frac{dv}{v} \\
&\geq d_W C_1 e^{-C_2} \int_{d(x, y)}^1 \frac{dv}{v} = d_W C_1 e^{-C_2} \log \frac{1}{d(x, y)}
\end{aligned}$$

as desired. $\square$

Particularly useful for the construction of Gaussian free fields on fractals in Section 3 is the following kernel expansion stated in [14, Proposition 2.4].

Lemma 3. *For any $s > d_S/2$, the fractional Riesz kernel (2.5) admits the expansion*

$$k_s(x, y) = \sum_{k=1}^{\infty} \lambda_k^{-s/2} \varphi_k(x) \varphi_k(y), \quad x, y \in K,$$

where the convergence of the series is understood in the $L^2(K \times K, \mu \otimes \mu)$ -sense.

Proof. For any $\ell \geq 1$, define

$$k_{s, \ell}(x, y) := \sum_{j=1}^{\ell} \lambda_j^{-s/2} \varphi_j(x) \varphi_j(y).$$

The orthogonality of the eigenfunctions φ_j together with Proposition 1 yield

$$\begin{aligned}
\|k_{s, \ell}(x, y)\|_{L^2(K \times K, \mu \otimes \mu)}^2 &= \sum_{j=1}^{\infty} \lambda_j^{-s} = \sum_{j=1}^{\infty} \sum_{k=N_{j-1}+1}^{N_j} \lambda_k^{-s} \\
&\simeq \sum_{j=1}^{\infty} (N_j - N_{j-1}) \Lambda_{j-1}^{-s} = \sum_{j=1}^{\infty} 3^j \Lambda_{j-1}^{-s} \simeq \sum_{j=1}^{\infty} 3^j 5^{-sj}
\end{aligned}$$

which is bounded for $s > \frac{d_S}{2} = \frac{\log 3}{\log 5}$. Here, $A \simeq B$ means that the quantity A is bounded above and below by different constants times the quantity B . $\square$

2.3. Function spaces. The function spaces playing a role in the current analysis are those associated with the Riesz potentials. In the sequel we follow its characterization from [14, Definition 3.1].

Definition 2. Let $s > 0$. The Sobolev-Riesz potential space of order s is defined as

$$H^s(K) := \{v = (-\Delta)^{-s/2} u \text{ for some } u \in L^2(K)\}$$

with associated norm given by

$$\|v\|_{H^s} := \|(-\Delta)^{s/2} v\|.$$

By virtue of the spectral decomposition, the norm above admits the expression

$$\|v\|_{H^s}^2 = \sum_{k=1}^{\infty} \lambda_k^s \langle v, \varphi_k \rangle^2. \quad (2.8)$$

Remark 2. We define the space $H^{-s}(K)$ as the completion of $L^2(K)$ with respect to the norm

$$\|v\|_{H^{-s}}^2 = \|(-\Delta)^{-s/2}v\|^2 = \sum_{k=1}^{\infty} \lambda_k^{-s} \langle v, \varphi_k \rangle^2,$$

so that, formally, $H^{-s}(K) = (-\Delta)^{s/2}[L^2(K)]$. Since

$$|\langle u, v \rangle| \leq \|u\|_{H^{-s}} \|v\|_{H^s}$$

for all $u \in L^2(K)$ and $v \in H^s(K)$, the inner product in L^2 continuously extends to a bilinear form

$$\langle \cdot, \cdot \rangle_{H^{-s}, H^s} : H^{-s}(K) \times H^s(K) \rightarrow \mathbb{R}.$$

In particular, $H^{-s}(K)$ can be identified with the dual space of $H^s(K)$. We simply write $\langle u, v \rangle$ for the extended dual pairing between $H^{-s}(K)$ and $H^s(K)$, whenever $u \in H^{-s}(K)$ and $v \in H^s(K)$.

Further, following [2] we consider the space of test functions

$$\mathcal{S}(K) := \left\{ u \in C_0(K) : \lim_{n \rightarrow \infty} n^k \left| \int_K \varphi_n(y) u(y) d\mu(y) \right| = 0 \quad \forall k \geq 0 \right\}$$

endowed with the topology induced by the family of norms $\{\|(-\Delta)^k u\|\}_{k \geq 0}$. We denote its dual by $\mathcal{S}'(K)$ and use the same notation $\langle \cdot, \cdot \rangle$ for the dual pairing between $\mathcal{S}(K)$ and $\mathcal{S}'(K)$. Note that $\mathcal{S}(K)$ embeds continuously into $H^s(K)$ for all $s > 0$.

3. FRACTAL GAUSSIAN FREE FIELDS

The aim of this section is to construct and analyze a random field h on K which may be viewed as the analogue to the planar Gaussian free field on the Sierpinski gasket. The law of this field is formally given by the “probability measure”

$$\text{Law}_h(du) = \frac{1}{Z} \exp \left(-\frac{1}{2} \|(-\Delta)^{d_S/4} u\|_{L^2}^2 \right) du$$

for a suitable (non-existing) normalization constant Z . Note that, due to the logarithmic divergence of the Riesz kernel k_{d_S} associated with $(-\Delta)^{-d_S/2}$, the field h cannot be defined pointwise, but only in a distributional sense.

Definition 3. We say that a centered Gaussian random element $h : \Omega \rightarrow \mathcal{S}'(K)$ is a *fractal Gaussian free field* (FGFF) on K if its covariance is given by

$$\mathbb{E}[\langle h, u \rangle \langle h, v \rangle] = \int_K \int_K k_{d_S}(x, y) u(x) v(y) d\mu(x) d\mu(y) \quad (3.1)$$

for all $u, v \in \mathcal{S}(K)$.

Theorem 4. *The FGFF on K exists, and it is unique in law.*

An abstract proof of the existence and uniqueness of h via the Bochner-Minlos theorem can be found in [2, Theorem 2.12]. It is also possible to provide a more constructive proof of the existence of the FGFF by means of a suitable approximating sequence of Gaussian fields, c.f. [2, Remark 2.4]. We analyze this approximation in the next section, which will further imply that $h \in H^{-\varepsilon}(K)$ $\mathbb{P}$ -a.s. for every $\varepsilon > 0$.

3.1. Approximating Gaussian fields. As in [2, Definition 2.16], let $\{\xi_k\}_{k \geq 1}$ be a sequence of i.i.d. standard normal distributed random variables. For each $\ell \geq 1$, define the random field

$$h_\ell(x) := \sum_{k=1}^{N_\ell} \xi_k \lambda_k^{-\frac{d_S}{4}} \varphi_k(x), \quad x \in K, \quad (3.2)$$

where N_ℓ is the multiplicity of the eigenvalue Λ_ℓ , see Proposition 1. The covariance function of this field is thus given by

$$k_{d_S, \ell}(x, y) := \mathbb{E}[h_\ell(x)h_\ell(y)] = \sum_{k=1}^{N_\ell} \lambda_k^{-\frac{d_S}{2}} \varphi_k(x)\varphi_k(y), \quad x, y \in K. \quad (3.3)$$

The convergence of the sequence $\{h_\ell\}_{\ell \geq 1}$ can be shown using the spectral properties of $-\Delta$ that were recorded in Proposition 1.

Theorem 5. (i) *The sequence $\{h_\ell\}_{\ell \geq 1}$ converges both in $L^2(\Omega)$ and $\mathbb{P}$ -a.s. to the FGFF h on K . In particular,*

$$h \stackrel{d}{=} \sum_{k=1}^{\infty} \xi_k \lambda_k^{-\frac{d_S}{4}} \varphi_k$$

and, $\mathbb{P}$ -a.s., it holds that $h \in H^{-\varepsilon}(K)$ for any $\varepsilon > 0$ and $h \notin L^2(K)$.

(ii) *For every $u \in H^{-d_S/2}(K)$, the sequence $\{\langle h_\ell, u \rangle\}_{\ell \geq 1}$ is a centered, L^2 -bounded martingale with respect to the filtration $\{\mathcal{F}_\ell\}_{\ell \geq 1}$, where*

$$\mathcal{F}_\ell := \sigma(\{\xi_1, \dots, \xi_{N_\ell}\}).$$

Moreover, the sequence converges in $L^2(\Omega)$ and $\mathbb{P}$ -a.s. to a centered Gaussian random variable, denoted by $h \cdot u$. In particular, $\mathbb{P}$ -a.s.

$$h \cdot u = \sum_{k=1}^{\infty} \xi_k \lambda_k^{-\frac{d_S}{4}} \langle \varphi_k, u \rangle$$

and

$$\mathbb{E}[(h \cdot u)^2] = \|u\|_{H^{-d_S/2}}^2.$$

Proof. (i) Fix $\varepsilon > 0$. We show first that $h \in H^{-\varepsilon}(K)$ $\mathbb{P}$ -a.s. By definition, applying (2.8) and Proposition 1 we have

$$\mathbb{E}[\|h\|_{H^{-\varepsilon}}^2] = \sum_{k=1}^{\infty} \lambda_k^{-\frac{d_S}{2}-\varepsilon} \mathbb{E}[\xi_k^2] = \sum_{j=1}^{\infty} \sum_{k=N_{j-1}+1}^{N_j} \lambda_k^{-\frac{d_S}{2}-\varepsilon} \simeq \sum_{j=1}^{\infty} (N_j - N_{j-1}) \Lambda_{j-1}^{-\frac{d_S}{2}-2\varepsilon}.$$

Once again, $A \simeq B$ means that the quantity A is bounded above and below by different constants times the quantity B . Substituting in the above the expression for N_j from (2.3),

$$\mathbb{E}[\|h\|_{H^{-\varepsilon}}^2] = \sum_{j=1}^{\infty} \left(\frac{1}{2}(3^{j+1} - 3) - \frac{1}{2}(3^j - 3) \right) \Lambda_{j-1}^{-\frac{d_S}{2}-\varepsilon}$$

$$= \sum_{j=1}^{\infty} 3^j \Lambda_{j-1}^{-\frac{d_S}{2}-\varepsilon} \simeq \sum_{j=1}^{\infty} 5^{-j\varepsilon}, \quad (3.4)$$

where the last line follows from Proposition 1. The series above is finite if and only if $\varepsilon > 0$, and the particular case $\varepsilon = 0$ yields $h \notin L^2(K)$ $\mathbb{P}$ -a.s.

Similarly, in view of (3.4) and (2.3), for any $\ell \geq 1$ it holds that

$$\mathbb{E}[\|h - h_\ell\|_{H^{-\varepsilon}}^2] \lesssim \sum_{j=N_\ell+1}^{\infty} 5^{-j\varepsilon} \xrightarrow{\ell \rightarrow \infty} 0$$

and thus $\{h_\ell\}_{\ell \geq 1}$ converges in $L^2(\Omega)$. By virtue of Tschebychev's inequality and (3.4), for any $\delta > 0$

$$\sum_{\ell=1}^{\infty} \mathbb{P}(\|h - h_\ell\|_{H^{-\varepsilon}} > \delta) \leq \frac{1}{\delta^2} \sum_{\ell=1}^{\infty} \mathbb{E}[\|h - h_\ell\|_{H^{-\varepsilon}}^2] < \infty.$$

The fast convergence theorem, see e.g. [16, Theorem 6.12], implies $\mathbb{P}$ -a.s. convergence and hence convergence in distribution.

Finally, we verify that h is in fact a FGFF: Since h is an $L^2(\Omega)$ -limit of centered Gaussian random variables, it is a centered Gaussian random variable itself with covariance

$$\begin{aligned} \mathbb{E}[\langle h, u \rangle \langle h, v \rangle] &= \sum_{k_1=1}^{\infty} \sum_{k_2=1}^{\infty} \mathbb{E} \left[\xi_{k_1} \xi_{k_2} (\lambda_{k_1} \lambda_{k_2})^{-\frac{d_S}{4}} \langle \varphi_{k_1}, u \rangle \langle \varphi_{k_2}, v \rangle \right] \\ &= \sum_{k=1}^{\infty} \lambda_k^{-\frac{d_S}{2}} \langle \varphi_k, u \rangle \langle \varphi_k, v \rangle \\ &= \int_K \int_K u(x) v(y) \sum_{k=1}^{\infty} \lambda_k^{-\frac{d_S}{2}} \varphi_k(x) \varphi_k(y) d\mu(y) d\mu(x) \\ &= \int_K \int_K u(x) v(y) k_{d_S}(x, y) d\mu(y) d\mu(x), \end{aligned}$$

for any $u, v \in H^{d_S/2}(K)$. Note that in the first equality we used the convergence in $L^2(\Omega)$ of h_ℓ to h , the independence of ξ_k in the second, and Lemma 3 in the last two lines.

(ii) Let $u \in H^{-d_S/2}(K)$. By definition of h_ℓ , for any $\ell \geq 1$

$$\langle h_\ell, u \rangle = \sum_{k=1}^{N_\ell} \xi_k \lambda_k^{-\frac{d_S}{4}} \langle \varphi_k, u \rangle,$$

which is a sum of independent centered (Gaussian) random variables and thus a martingale with respect to the natural filtration $\mathcal{F}_\ell$. Moreover, independence implies that

$$\begin{aligned} \sup_{\ell \in \mathbb{N}} \mathbb{E}[|\langle h_\ell, u \rangle|^2] &= \sup_{\ell \in \mathbb{N}} \mathbb{E}\left[\left|\sum_{k=1}^{N_\ell} \xi_k \lambda_k^{-\frac{d_S}{4}} \langle \varphi_k, u \rangle\right|^2\right] \\ &= \sup_{\ell \in \mathbb{N}} \sum_{k=1}^{N_\ell} \mathbb{E}[\xi_k^2] \lambda_k^{-\frac{d_S}{2}} |\langle \varphi_k, u \rangle|^2 \\ &= \sum_{k=1}^{\infty} \lambda_k^{-\frac{d_S}{2}} |\langle \varphi_k, u \rangle|^2 = \|u\|_{H^{-d_S/2}}^2 < \infty. \end{aligned}$$

Therefore, $\{\langle h_\ell, u \rangle\}_{\ell \geq 1}$ is an $L^2(\Omega)$ -bounded martingale. By virtue of Doob's L^p -convergence theorem for martingales, see e.g. [16, Theorem 11.10], the sequence converges $\mathbb{P}$ -a.s. and in $L^2(\Omega)$ to a random variable denoted by $h \cdot u \in L^2(\Omega)$. Moreover,

$$h \cdot u = \sum_{k=1}^{\infty} \xi_k \lambda_k^{-\frac{d_S}{4}} \langle \varphi_k, u \rangle \quad \mathbb{P}\text{-a.s.}$$

Since each $\langle h_\ell, u \rangle$ defines a centered Gaussian random variable, the L^2 -convergence of $\langle h_\ell, u \rangle$ to $h \cdot u$ implies that $h \cdot u$ is a centered Gaussian random variable with variance given by $\|u\|_{H^{-d_S/2}}^2$. □

Remark 3. Analogous arguments as those in the proof of Theorem 5(ii) yield that the FGFF h can always be approximated by the sequence of Gaussian random fields

$$h_\ell(x) := \sum_{k=1}^{N_\ell} \tilde{\xi}_k \lambda_k^{-\frac{d_S}{4}} \varphi_k(x), \quad x \in K,$$

with $\tilde{\xi}_k := \langle h, \varphi_k \rangle$.

We now discuss how the FGFF interplays with the symmetries and scaling properties of K .

3.2. Invariance properties.

3.2.1. *Invariance by symmetries.* Let $\sigma_i: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $i = 1, 2, 3$, denote each of the reflections about the axes that divide the Sierpinski gasket into half, see Figure 2 .

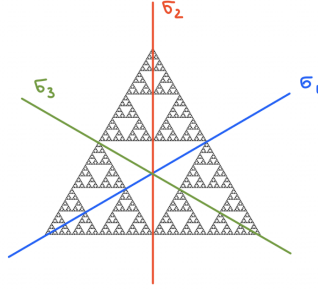


FIGURE 2. The reflections σ_1 , σ_2 and σ_3 .

Then, $\sigma_i(V_0) = V_0$ and

$$\mathcal{E}(u \circ \sigma_i, v \circ \sigma_i) = \mathcal{E}(u, v).$$

In particular, as in the proof of [3, Proposition 3.9], for any $i = 1, 2, 3$ and $s > 0$ it holds that

$$k_s(\sigma_i(x), \sigma_i(y)) = k_s(x, y). \quad (3.5)$$

Since the underlying measure μ is also invariant under the symmetries σ_i , the invariance of the FGFF on K follows.

Theorem 6. *For any $i = 1, 2, 3$ it holds that*

$$h \stackrel{d}{=} h \circ (\sigma_i)^{-1}, \quad (3.6)$$

where $h \circ (\sigma_i)^{-1}$ is understood in the distributional sense, i.e. $\langle h \circ (\sigma_i)^{-1}, u \rangle = \langle h, u \circ \sigma_i \rangle$.

Proof. By construction, both h and $h \circ (\sigma_i)^{-1}$ are centered Gaussian fields. Moreover, by the invariance (3.5) of the kernel k_{d_S} and that of the measure μ , they have the same covariance function, which implies (3.6). $\square$

3.2.2. Self-similarity by scaling. We prove now the self-similarity of the FGFF on K under the contractions F_w from Section 2.1, where $w = w_1 w_2 \dots w_n \in \{1, 2, 3\}^n$, $n \geq 1$. For a fixed word w , let $K_w := F_w(K) \subset K$ denote the rescaled Sierpinski gasket associated with the contraction mapping F_w . The space is naturally equipped with the rescaled metric d_w and measure μ_w that satisfy

$$\begin{aligned} d_w(x, y) &= 2^{-n} d(F_w^{-1}(x), F_w^{-1}(y)), \\ (F_w^{-1})_{\#} \mu_w &= 3^{-n} \mu. \end{aligned} \quad (3.7)$$

In particular, for any $u, v \in L^2(K_w, \mu_w)$

$$\langle u, v \rangle_{L^2(K_w, \mu_w)} = 3^{-n} \langle u \circ F_w, v \circ F_w \rangle_{L^2(K, \mu)}.$$

The standard Dirichlet form with Dirichlet boundary conditions $(\mathcal{E}_w, \mathcal{F}_w)$ on $L^2(K_w, \mu_w)$ satisfies

$$\mathcal{E}_w(u, v) = \left(\frac{5}{3}\right)^n \mathcal{E}(u \circ F_w, v \circ F_w)$$

and note that $u|_{F_w(V_0)} \equiv 0$ for all $u \in \mathcal{F}_w$. As in the proof of [3, Proposition 3.11], the associated heat kernel satisfies

$$p_t^w(x, y) = 3^n p_{5^n t}(F_w^{-1}(x), F_w^{-1}(y)) \quad (3.8)$$

for all $x, y \in K_w$, see e.g. [3, Section 3.4.2].

To state the self-similarity under F_w properly, we would like to consider a random field h^w on K_w formally given by

$$h^w = h \circ F_w^{-1}.$$

Note however, that the above equality has to be read in a distributional sense because h is not defined pointwise. Therefore, analogously to Remark 2, by regarding $H^{-\varepsilon}(K_w)$ as the completion of $L^2(K_w, \mu_w)$ with respect to the corresponding norm $\|\cdot\|_{H^{-\varepsilon}(K_w)}$ we set

$$\langle h^w, u \rangle_{H^{-\varepsilon}(K_w), H^{\varepsilon}(K_w)} := 3^{-n} \langle h, u \circ F_w \rangle_{H^{-\varepsilon}(K), H^{\varepsilon}(K)} \quad (3.9)$$

for any $u \in H^{\varepsilon}(K_w)$. In this way we may now state and prove the corresponding self-similarity of the FGFF h .

Theorem 7. *Let h be the FGFF on K . For any $w \in \{1, 2, 3\}^n$ and $n \geq 1$,*

$$h^w := h \circ F_w^{-1},$$

defined in the distributional sense (3.9), is the FGFF on K_w .

Proof. Note first that, by definition, h^w is a centered Gaussian field on K_w . In view of (3.8) and the definition of the Riesz kernel (2.5) it follows that

$$k_s^w(F_w(x), F_w(y)) = \frac{3^n}{5^{ns/2}\Gamma(s/2)} \int_0^\infty t^{s/2-1} p_t(x, y) dt = \frac{3^n}{5^{ns/2}} k_s(x, y)$$

for μ -a.e. $x, y \in K$. In particular, in the critical case $s = d_S$ the latter reads

$$k_{d_S}^w(F_w(x), F_w(y)) = k_{d_S}(x, y). \quad (3.10)$$

In addition, for any $u, v \in H^\varepsilon(K_w)$

$$\begin{aligned} \mathbb{E}[\langle h^w, u \rangle \langle h^w, v \rangle] &= 3^{-2n} \mathbb{E}[\langle h, u \circ F_w \rangle \langle h, v \circ F_w \rangle] \\ &= 3^{-2n} \int_K \int_K k_{d_S}(x, y) u(F_w(x)) v(F_w(y)) d\mu(x) d\mu(y). \end{aligned}$$

By performing a change of variables, using (3.10) and (3.7), we finally obtain

$$\begin{aligned} \mathbb{E}[\langle h^w, u \rangle \langle h^w, v \rangle] &= 3^{-2n} \int_K \int_K k_{d_S}^w(F_w(x), F_w(y)) u(F_w(x)) v(F_w(y)) d\mu(x) d\mu(y) \\ &= \int_{K_w} \int_{K_w} k_{d_S}^w(\tilde{x}, \tilde{y}) u(\tilde{x}) v(\tilde{y}) d\mu_w(\tilde{x}) d\mu_w(\tilde{y}). \end{aligned}$$

□

4. LIOUVILLE QUANTUM GRAVITY MEASURE

The aim of this section is to construct a random measure that takes into account the geometry induced by the FGFF h on K . Formally, such a measure is given by

$$e^{\gamma h(x)} d\mu(x)$$

for some suitable $\gamma \in \mathbb{R}$. However, since h is almost surely not a function, we need to replace h by appropriately rescaled and well-defined approximations.

In the special case of the Gaussian free field in $\mathbb{R}^2$, the resulting random measure is called *Liouville Quantum Gravity measure* (LQG measure). The concept appeared in theoretical physics in the early 1980s, see [19], and its rigorous probabilistic construction has been treated afterwards by several authors [15, 9, 20, 4, 22].

Remark 4. Throughout this section, the parameter γ is chosen in the range

$$|\gamma| < \sqrt{d_H/C_0},$$

where $C_0 > 0$ is the constant from (2.6).

4.1. Approximations. In this section we analyze the approximating measures that arise from the eigenfunction approximation h_ℓ with covariance kernels $k_{d_S, \ell}$ introduced in (3.2) and (3.3) respectively. For fixed $\gamma > 0$ and $\ell \geq 1$, these measures are defined as

$$d\nu_\ell^\gamma(x) := e^{\gamma h_\ell(x) - \frac{\gamma^2}{2} \mathbb{E}[h_\ell(x)^2]} d\mu(x). \quad (4.1)$$

Note that h_ℓ is by construction a (centered) Gaussian field, whence the distribution of its is fully determined by its covariance structure. In particular, we show that $\nu_\ell^\gamma(K)$ is a non-negative $L^2(\Omega)$ -bounded martingale for $|\gamma| < \sqrt{d_H/C_0}$.

Proposition 8. *Let $|\gamma| < \sqrt{d_H/C_0}$. For any $u \in C_b(K)$, the sequence $\{\nu_\ell^\gamma(u)\}_{\ell \geq 1}$ given by*

$$\nu_\ell^\gamma(u) := \int_K u d\nu_\ell^\gamma, \quad \ell \geq 1,$$

is an $L^2(\Omega)$ -bounded martingale with respect to the filtration $\mathcal{F} = \{\mathcal{F}_\ell\}_{\ell \geq 1}$, where

$$\mathcal{F}_\ell := \sigma(h_\ell), \quad \ell \geq 1.$$

Proof. Let $u \in C_b(K)$. We first show that $\nu_\ell^\gamma(u) \in L^1(\Omega)$ for all $\ell \geq 1$: Since $h_\ell(x)$ is a centered Gaussian random variable with variance $k_{d_S, \ell}(x, x)$, its moment generating function equals

$$\mathbb{E}[e^{\gamma h_\ell(x)}] = e^{\frac{\gamma^2}{2} \mathbb{E}[h_\ell^2(x)]} = e^{\frac{\gamma^2}{2} k_{d_S, \ell}(x, x)}, \quad (4.2)$$

where the last equality follows from (3.3). Thus,

$$\mathbb{E}[|\nu_\ell^\gamma(u)|] \leq \int_K |u(x)| \mathbb{E}[e^{\gamma h_\ell(x)}] e^{-\frac{\gamma^2}{2} k_{d_S, \ell}(x, x)} d\mu(x) = \int_K |u| d\mu < \infty.$$

To show the martingale property, we rewrite

$$d\nu_{\ell+1}^\gamma(u)(x) = Y_{\ell, \ell+1}(x) d\nu_\ell^\gamma(u)(x),$$

with

$$Y_{\ell, \ell+1}(x) = e^{\gamma(h_{\ell+1} - h_\ell)(x) - \frac{\gamma^2}{2} (k_{d_S, \ell+1} - k_{d_S, \ell})(x, x)}.$$

Note now that

$$\mathbb{E}[Y_{\ell, \ell+1}(x) | \mathcal{F}_\ell] = \mathbb{E}[Y_{\ell, \ell+1}(x)] = 1,$$

where the last equality follows analogously to (4.2). Thus,

$$\begin{aligned} \mathbb{E}[\nu_{\ell+1}^\gamma(u) | \nu_\ell^\gamma(u)] &= \int_K u(x) \mathbb{E}[Y_{\ell, \ell+1}(x) | \nu_\ell^\gamma(u)] d\nu_\ell^\gamma(u)(x) \\ &= \int_K u(x) d\nu_\ell^\gamma(u)(x) = \nu_\ell^\gamma(u), \end{aligned}$$

whence $\{\nu_\ell^\gamma(u)\}_{\ell \geq 1}$ is a martingale.

To show the $L^2(\Omega)$ -boundedness, we prove that

$$\sup_{\ell \geq 1} \mathbb{E}[\nu_\ell^\gamma(u)^2] < \infty.$$

Applying Fubini, (3.1) and (4.2), we obtain

$$\begin{aligned}
\mathbb{E}[\nu_\ell^\gamma(u)^2] &= \mathbb{E}\left[\left(\int_K u d\nu_\ell^\gamma\right)^2\right] \\
&= \mathbb{E}\left[\int_K \int_K u(x)u(y)e^{\gamma(h_\ell(x)+h_\ell(y))-\frac{\gamma^2}{2}\mathbb{E}[h_\ell(x)^2+h_\ell(y)^2]}d\mu(y)d\mu(x)\right] \\
&= \int_K \int_K u(x)u(y)e^{-\frac{\gamma^2}{2}\mathbb{E}[h_\ell(x)^2+h_\ell(y)^2]}\mathbb{E}\left[e^{\gamma(h_\ell(x)+h_\ell(y))}\right]d\mu(y)d\mu(x) \\
&= \int_K \int_K u(x)u(y)e^{-\frac{\gamma^2}{2}(k_{d_s,\ell}(x,x)+k_{d_s,\ell}(y,y))}\mathbb{E}\left[e^{\gamma(h_\ell(x)+h_\ell(y))}\right]d\mu(y)d\mu(x) \\
&= \int_K \int_K u(x)u(y)e^{-\frac{\gamma^2}{2}(k_{d_s,\ell}(x,x)+k_{d_s,\ell}(y,y))+\frac{\gamma^2}{2}(k_{d_s,\ell}(x,x)+k_{d_s,\ell}(y,y)+2k_{d_s,\ell}(x,y))}d\mu(y)d\mu(x) \\
&= \int_K \int_K u(x)u(y)e^{\gamma^2 k_{d_s,\ell}(x,y)}d\mu(y)d\mu(x).
\end{aligned}$$

Since u is bounded and K is compact, the above estimate, Proposition 2 and Lemma 9 below finally yield

$$\begin{aligned}
\mathbb{E}\left[\left(\int_K u d\nu_\ell^\gamma\right)^2\right] &\leq \|u\|_{L^\infty}^2 \int_K \int_K e^{\gamma^2 |k_{d_s,\ell}(x,y)|}d\mu(y)d\mu(x) \\
&\leq \|u\|_{L^\infty}^2 \int_K \int_K e^{\gamma^2 C(1-\log d(x,y))}d\mu(y)d\mu(x) \\
&= \|u\|_{L^\infty}^2 e^{\gamma^2 C} \int_K \int_K \frac{1}{d(x,y)^{\gamma^2 C}}d\mu(y)d\mu(x) \\
&\leq \tilde{C}\|u\|_{L^\infty}^2.
\end{aligned}$$

□

The following estimate used in the previous proof appears in [2] and its proof is included here for completeness.

Lemma 9. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then,*

$$\int_K \int_K \frac{1}{d(x,y)^{\gamma^2 C}}d\mu(y)d\mu(x) < \infty.$$

Proof. For any $r > 0$ and $x \in K$, let $B(x, r)$ denote the ball of radius r centered at x . Using a standard integral decomposition argument and the d_H -Ahlfors regularity of μ ,

$$\begin{aligned}
\int_K \int_K \frac{1}{d(x,y)^{\gamma^2 C_0}}d\mu(y)d\mu(x) &\leq \int_K \sum_{k=0}^{\infty} \int_{B(x, 2^{-k}R_K) \setminus B(x, 2^{-k-1}R_K)} \frac{1}{d(x,y)^{\gamma^2 C_0}}d\mu(y)d\mu(x) \\
&\leq \int_K \sum_{k=0}^{\infty} \int_{B(x, 2^{-k}) \setminus B(x, 2^{-k-1})} (2^{k+1})^{\gamma^2 C_0}d\mu(y)d\mu(x) \\
&\leq \int_K \sum_{k=0}^{\infty} (2^{k+1})^{\gamma^2 C_0} \mu(B(x, 2^{-k}))d\mu(x) \\
&\lesssim \int_K \sum_{k=0}^{\infty} (2^{k+1})^{\gamma^2 C_0} (2^{-k})^{d_H}d\mu(x)
\end{aligned}$$

$$\lesssim \mu(K) 2^{\gamma^2 C} \sum_{k=0}^{\infty} 2^{-k(d_H - \gamma^2 C_0)}.$$

The latter series converges if and only if $d_H - \gamma^2 C_0 > 0$, which is the case since by assumption $|\gamma| < \sqrt{d_H/C_0}$. $\square$

4.2. Liouville Quantum Gravity measure. Next, we give the definition of the Liouville quantum gravity measure. Here, we use the characterization through the transformation rule (4.3) established in [22].

Definition 4. We call a random measure ν^γ on K a *Liouville Quantum Gravity measure* (LQG measure) if

$$\mathbb{E} \left[\int_K F(h, x) d\nu^\gamma(x) \right] = \mathbb{E} \left[\int_K F(h + \gamma k_{d_S}(x, \cdot), x) d\mu(x) \right] \quad (4.3)$$

for every nonnegative measurable functional F on $H^{-\varepsilon}(K) \times K$.

Theorem 10. For any $|\gamma| < \sqrt{d_H/C_0}$, there exists a unique LQG measure ν^γ on K such that

$$\mathbb{E}[\nu^\gamma(E)] = \mu(E), \quad (4.4)$$

for any $E \subseteq K$. In addition, ν_ℓ^γ converges weakly to ν^γ in the sense of Borel measures on K in $L^2(\Omega)$, i.e.

$$\nu_\ell^\gamma(u) \xrightarrow[\ell \rightarrow \infty]{L^2(\Omega)} \nu^\gamma(u)$$

for all $u \in C_b(K)$.

Proof. The uniqueness follows immediately from [22, Corollary 5].

The existence follows by approximation: By Proposition 8, the sequence $\{\nu_\ell^\gamma(u)\}_{\ell \geq 1}$ is an L^2 -bounded martingale for any $u \in C_b(K)$. The L^p -convergence theorem for martingales (with $p = 2$), see e.g. [16, Theorem 11.4], guarantees the existence of a limit in $L^2(\Omega)$. The convergence in the sense of Borel measures follows then by the Riesz-Markov-Kakutani representation theorem.

Moreover, for each $\ell \geq 1$ and in view of (4.1), the measure ν_ℓ^γ is absolutely continuous with respect to μ and has Radon-Nikodym derivative $e^{\gamma h_\ell(x) - \frac{\gamma^2}{2} k_{d_S, \ell}(x, x)}$ for μ -a.e. $x \in K$. By the Cameron-Martin theorem, the Radon-Nikodym derivative is precisely the density of the law of the shifted field $h_\ell + \gamma k_{d_S, \ell}(x, \cdot)$ and thus

$$\mathbb{E} \left[\int_K F(h_\ell, x) d\nu_\ell^\gamma(x) \right] = \mathbb{E} \left[\int_K F(h_\ell + \gamma k_{d_S, \ell}(x, \cdot), x) d\mu(x) \right]$$

for any positive measurable functional F on $H^{-\varepsilon}(K) \times K$. In particular, see [22, Theorem 4], each ν_ℓ^γ is a subcritical GMC with expectation μ . Since in addition the sequence $\{\nu_\ell^\gamma(K)\}_{\ell \geq 1}$ is uniformly integrable, it follows from [22, Theorem 3] that ν^γ is also a GMC, which by [22, Theorem 4] satisfies (4.3). $\square$

Remark 5. In the proof of Theorem 10 we rely on the fact that $\{\nu_\ell^\gamma(K)\}_{\ell \geq 1}$ is an L^2 -bounded martingale for $|\gamma| < \sqrt{d_H/C_0}$. Alternatively, one could prove $\sup_\ell \mathbb{E}[\nu_\ell^\gamma(K)^p] < \infty$ for some $p > 1$ depending on γ . This would imply uniform integrability and hence convergence by Doob's convergence theorem. Such a moment estimate could potentially

extend the domain of admissible values for the parameter γ . A standard way of proving L^p -bounds is by controlling the contribution of so-called *thick points*, see [9, 4]. However, these arguments require precise logarithmic asymptotics of the kernel, which we do not have in this case as the available heat kernel bounds on K only yield two-sided bounds with different constants.

We record some properties of the LQG measure ν^γ .

Corollary 11. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then ν^γ is $\mathbb{P}$ -a.s. finite and non-trivial.*

Proof. Since (4.4) implies $\mathbb{E}[\nu^\gamma(K)] < \infty$, it follows $\nu^\gamma(K) < \infty$ $\mathbb{P}$ -a.s. To show that ν^γ is $\mathbb{P}$ -a.s. non-trivial, we first note that

$$\mathbb{P}(\nu^\gamma(K) > 0) > 0.$$

Actually, (4.4) gives $\mathbb{E}[\nu^\gamma(K)] = \mu(K) > 0$. Since $\nu^\gamma(K) \geq 0$ $\mathbb{P}$ -a.s., the probability of the event $\{\nu^\gamma(K) > 0\}$ must be positive. The latter probability equals one due to Kolmogorov's zero-one law. We shall explain the argument in more generality in the proof of Corollary 12 below. $\square$

In fact, one can show that ν^γ has $\mathbb{P}$ -a.s. full support. The proof uses Kolmogorov's zero-one law, which is a standard argument in Gaussian multiplicative chaos, see e.g. [20, Section 2].

Corollary 12. *For any $|\gamma| < \sqrt{d_H/C_0}$, the measure ν^γ has $\mathbb{P}$ -a.s. full support.*

Proof. We first note that (4.4) implies $\mathbb{P}(\nu^\gamma(A) > 0) > 0$ for every non-empty open set $A \subseteq K$ since $\mu(A) > 0$ and ν^γ is non-negative $\mathbb{P}$ -a.s. Now, let us define for any $1 \leq \ell \leq m$ the random measure

$$d\mu_{m,\ell}^\gamma(x) := Y_{m,\ell}(x) dx,$$

where

$$Y_{m,\ell}(x) := e^{\gamma(h_m - h_\ell)(x) - \frac{\gamma^2}{2}(k_{d_S,m} - k_{d_S,\ell})(x,x)}.$$

By virtue of Theorem 10, for each fixed $\ell \geq 1$, the sequence $\{\mu_{m,\ell}^\gamma\}_{m \geq \ell}$ converges to a random measure μ_ℓ^γ as $m \rightarrow \infty$ and

$$\inf_{x \in A} e^{\gamma h_\ell(x) - \frac{\gamma^2}{2} k_{d_S,\ell}(x,x)} \mu_\ell^\gamma(A) \leq \nu^\gamma(A) \leq \sup_{x \in A} e^{\gamma h_\ell(x) - \frac{\gamma^2}{2} k_{d_S,\ell}(x,x)} \mu_\ell^\gamma(A).$$

In particular, $\{\nu^\gamma(A) > 0\} = \{\mu_\ell^\gamma(A) > 0\}$ for all $\ell \geq 1$. Moreover, since

$$(h_m - h_\ell)(x) = \sum_{k=N_{\ell+1}}^{N_m} \xi_k \lambda_k^{-\frac{d_S}{4}} \varphi_k(x)$$

for every $m \geq \ell$, the random measure μ_ℓ^γ only depends on the random variables $(\xi_{N_{\ell+1}}, \xi_{N_{\ell+2}}, \dots)$. Thus,

$$\{\mu_\ell^\gamma(A) > 0\} \in \sigma(\xi_{N_{\ell+1}}, \xi_{N_{\ell+2}}, \dots)$$

and hence

$$\{\nu^\gamma(A) > 0\} \in \sigma(\xi_{N_{\ell+1}}, \xi_{N_{\ell+2}}, \dots)$$

for all $\ell \geq 1$. Consequently,

$$\{\nu^\gamma(A) > 0\} \in \bigcap_{\ell \geq 1} \sigma(\xi_{N_{\ell+1}}, \xi_{N_{\ell+2}}, \dots),$$

which is the tail sigma-field of the independent sequence $\{\xi_k\}_{k \geq 1}$. By Kolmogorov's zero-one law,

$$\mathbb{P}(\nu^\gamma(A) > 0) \in \{0, 1\}$$

and since we already know $\mathbb{P}(\nu^\gamma(A) > 0) > 0$, the above probability must equal one. Thus, ν^γ has full support $\mathbb{P}$ -a.s. as claimed. $\square$

The next corollary in particular implies that ν^γ has no atoms.

Corollary 13. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then, for any $a < d_H - C_0\gamma^2$,*

$$\mathbb{E} \left[\int_K \int_K \frac{1}{d(x, y)^a} d\nu^\gamma(x) d\nu^\gamma(y) \right] < \infty.$$

In particular, any $E \subseteq K$ with $\dim_H(E) \leq d_H - C\gamma^2$ satisfies $\nu^\gamma(E) = 0$ $\mathbb{P}$ -a.s.

Proof. By the same arguments as in the proof of Proposition 8,

$$\sup_{\ell \in \mathbb{N}} \mathbb{E} \left[\int_K \int_K \frac{1}{d(x, y)^a} d\nu_\ell^\gamma(x) d\nu_\ell^\gamma(y) \right] \leq \int_K \int_K \frac{1}{d(x, y)^{a+\gamma^2 C}} d\mu(x) d\mu(y),$$

where the right-hand side is finite due to Lemma 9. Then, Portmanteau's theorem implies

$$\mathbb{E} \left[\int_K \int_K \frac{1}{d(x, y)^a} d\nu^\gamma(x) d\nu^\gamma(y) \right] \leq \liminf_{\ell \rightarrow \infty} \mathbb{E} \left[\int_K \int_K \frac{1}{d(x, y)^a} d\nu_\ell^\gamma(x) d\nu_\ell^\gamma(y) \right] < \infty,$$

which yields

$$\int_K \int_K \frac{1}{d(x, y)^a} d\nu^\gamma(x) d\nu^\gamma(y) < \infty \quad \mathbb{P}\text{-a.s.}$$

Finally, Frostman's lemma [10, Theorem 4.13] implies that $\nu^\gamma(E) = 0$ for any $E \subseteq K$ such that the Hausdorff dimension is at most $d_H - C_0\gamma^2$. $\square$

4.3. Invariance properties.

4.3.1. Invariance by symmetries. We analyze the interplay of the LQG measure on K and the symmetries σ_1 , σ_2 and σ_3 described in Section 3.2.1. In the sequel, given a measure ν and a map F on K we denote by $F_\# \nu$ the push-forward of ν by F .

Theorem 14. *Let $|\gamma| < \sqrt{d_H/C_0}$ and let ν^γ be the LQG measure on K . For any $i = 1, 2, 3$,*

$$\nu^\gamma \stackrel{d}{=} (\sigma_i)_\# \nu^\gamma.$$

Proof. Note that ν^γ is the LQG measure associated to the FGFF h , while $(\sigma_i)_\# \nu^\gamma$ is the LQG measure associated to $h \circ (\sigma_i)^{-1}$. Since $h \stackrel{d}{=} h \circ (\sigma_i)^{-1}$ by Theorem 6, the assertion follows. $\square$

4.3.2. Self-similarity by scaling. For each word $w \in \{1, 2, 3\}^n$, $n \geq 1$, let ν_w^γ denote the LQG measure associated with the FGFF h^w on K^w defined in Theorem 7. We show next that, in distribution, the measure ν_w^γ only depends of the length of w .

Theorem 15. *Let $|\gamma| < \sqrt{d_H/C_0}$. For any $w \in \{1, 2, 3\}^n$, $n \geq 1$, the LQG measures ν^γ and ν_w^γ satisfy*

$$\nu_w^\gamma \stackrel{d}{=} 3^{-n} (F_w)_\# \nu^\gamma. \quad (4.5)$$

Proof. Let $w \in \{1, 2, 3\}^n$, $n \geq 1$. According to Theorem 7, $h_w \stackrel{d}{=} h \circ (F_w^{-1})$. Further, set

$$h_\ell^w := h_\ell \circ (F_w^{-1}), \quad (4.6)$$

where h_ℓ is the approximate FGFF from (3.2). By virtue of Theorem 5, $\{h_\ell\}_{\ell \geq 1}$ converges $\mathbb{P}$ -a.s. and in $L^2(\Omega)$ to the FGFF h on K , and analogously $\{h_\ell^w\}_{\ell \geq 1}$ converges to h^w . For each $\ell \geq 1$, the covariance kernel associated with h_ℓ^w is given by

$$k_\ell^w(\tilde{x}, \tilde{y}) := \mathbb{E}[h_\ell^w(\tilde{x})h_\ell^w(\tilde{y})] = k_{d_s, \ell}(F_w^{-1}(\tilde{x}), F_w^{-1}(\tilde{y})) \quad (4.7)$$

for all $\tilde{x}, \tilde{y} \in K_w$. Analogous arguments as those used in Theorem 10 imply that the associated LQG measure $\nu_{w, \ell}^\gamma$ converges to the LQG measure ν_w^γ on K_w , i.e.

$$\int_{K_w} u d\nu_{w, \ell}^\gamma \xrightarrow{L^1(\Omega)} \int_{K_w} u d\nu_w^\gamma$$

holds for all $u \in C_b(K_w)$.

We now show the invariance (4.5): For any $u \in C_b(K_w)$, it follows from (3.7), (4.6) and (4.7) that

$$\begin{aligned} \int_{K_w} u(\tilde{x}) d\nu_w^\gamma(\tilde{x}) &= \lim_{\ell \rightarrow \infty} \int_{K_w} u(\tilde{x}) e^{\gamma h_\ell^w(\tilde{x}) - \frac{\gamma^2}{2} k_{d_s, \ell}^w(\tilde{x}, \tilde{x})} d\mu_w(\tilde{x}) \\ &= 3^{-n} \lim_{\ell \rightarrow \infty} \int_K u(F_w(x)) e^{\gamma h_\ell^w(F_w(x)) - \frac{\gamma^2}{2} k_{d_s, \ell}^w(F_w(x), F_w(x))} d\mu(x) \\ &\stackrel{d}{=} 3^{-n} \lim_{n \rightarrow \infty} \int_K u(F_w(x)) e^{\gamma h_\ell(x) - \frac{\gamma^2}{2} k_{d_s, \ell}(x, x)} d\mu(x) \\ &= 3^{-n} \int_K u(F_w(x)) d\nu^\gamma(x) \end{aligned}$$

as we wanted to prove. $\square$

5. LIOUVILLE BROWNIAN MOTION

Similarly to the case of planar Liouville Brownian motion (LBM) introduced by Garban-Rhodes-Vargas in [13], LBM on the Sierpinski gasket will arise as a process defined through a time change of the (standard) Brownian motion associated with the Dirichlet form (2.1).

More precisely, let ρ be an initial probability distribution on K and let $\{B_t\}_{t \geq 0}$ denote the Hunt process on $(\Omega, \mathbb{Q}_\rho)$ associated with the standard Dirichlet form $(\mathcal{E}, \mathcal{F})$ from (2.1).

The LBM on K constructed in this section may be regarded as the diffusion process $\{B_t^\gamma\}_{t \geq 0}$ under the LQG measure ν^γ analyzed in Section 4.2. Formally, the process arises as the solution to the SDE

$$dB_t^\gamma = e^{-\frac{\gamma}{2} h(B_t^\gamma) + \frac{\gamma^2}{4} \mathbb{E}[h(B_t^\gamma)^2]} dB_t.$$

By Dubins-Schwarz, the associated time change is given by

$$A_t^\gamma = \int_0^t e^{\gamma h(B_s) - \frac{\gamma^2}{2} \mathbb{E}[h(B_s)^2]} ds.$$

The latter is a positive continuous additive functional (PCAF) which will be shown to be in *Revuz correspondence* with the measure ν^γ , i.e.

$$\mathbb{E}_\rho \left[\int_0^t u(B_s^\gamma) dA_s^\gamma \right] = \int_0^t \int_K \int_K u(y) p_s(x, y) d\rho(x) d\nu^\gamma(y) ds,$$

c.f. Theorem 19 and [12, Section 5.1] for definitions and details concerning PCAFs and smooth measures.

5.1. Capacities and smooth measures. To prove the existence of the process $\{B_t^\gamma\}_{t \geq 0}$ we rely on general Dirichlet form theory and first show that the LQG measure ν^γ is $\mathbb{P}$ -a.s. *smooth*, i.e. it is a positive Borel measure that charges no sets of $\mathcal{E}_1$ -capacity zero and admits a generalized nest, see [12, Chapter 2.2]. The latter property follows from the fact that K is compact and ν^γ is a finite measure. Thus, it only remains to verify whether it charges no sets of $\mathcal{E}_1$ -capacity zero.

To this end, recall e.g. from [12, Section 2.1] the symmetric form $(\mathcal{E}_1, \mathcal{F})$ on $L^2(K)$ given by

$$\mathcal{E}_1(u, v) := \mathcal{E}(u, v) + \langle u, v \rangle$$

for any $u, v \in \mathcal{F}$. The $\mathcal{E}_1$ -capacity of an open set $A \subset K$ is now defined as

$$\text{cap}(A) := \inf\{\mathcal{E}_1(u, u) : u \in \mathcal{F}, u \geq 1 \text{ } \mu\text{-a.e. on } A\}, \quad (5.1)$$

where we set $\inf \emptyset := \infty$. The $\mathcal{E}_1$ -capacity of a generic set $E \subset K$ is then given by

$$\text{cap}(E) := \inf_{B \text{ open}, E \subset B} \text{cap}(B),$$

and a set E with $\text{cap}(E) = 0$ is called an $\mathcal{E}_1$ -polar set.

Remark 6. Here, we use the convention of $\mathcal{E}_1$ -capacity, which is used in [12, Section 2.1]. Since $\mathcal{E}(u, u) \geq \lambda_1 \|u\|_{L^2}^2$ due to the positivity of the first Dirichlet eigenvalue of the Laplacian, $\mathcal{E}$ and $\mathcal{E}_1$ determine equivalent norms on $\mathcal{F}$. In particular, $\mathcal{E}$ -polar sets coincide with $\mathcal{E}_1$ -polar sets.

In order to prove that the measure ν^γ charges no $\mathcal{E}_1$ -polar sets, i.e. that for $E \subset K$

$$\text{cap}(E) = 0 \quad \Rightarrow \quad \nu^\gamma(E) = 0 \quad \mathbb{P}\text{-a.s.},$$

we study the related *Bessel potentials*. By analogy with [14, Definition 2.3], for $s > 0$ and $\alpha \geq 0$ we define the Bessel kernel

$$r_{s,\alpha}(x, y) := \frac{1}{\Gamma(s)} \int_0^\infty e^{-\alpha t} t^{s-1} p_t(x, y) dt, \quad x, y \in K, x \neq y,$$

and its associated Bessel potential

$$R_{s,\alpha} f(x) := \int_K r_{s,\alpha}(x, y) f(y) d\mu(y),$$

where $f \in L^2(K)$. Similarly to [14, Proposition 2.4] we may write the latter in terms of the infinitesimal generator Δ as

$$R_{s,\alpha} f = (\alpha I - \Delta)^{-s} f,$$

for any $f \in L^2(K)$. For $\nu \in \mathcal{M}_b(K)$, we further set

$$R_{s,\alpha} \nu(x) := \int_K r_{s,\alpha}(x, y) d\nu(y).$$

Lemma 16. *For any $E \subseteq K$, let*

$$b(E) := \sup\{\nu(E) : \nu \in \mathcal{M}_b(K), \|R_{1/2,1}(\mathbf{1}_E \nu)\|_{L^2} \leq 1\}.$$

Then, for every open set $A \subset K$ it holds that

$$\text{cap}(A) = b(A)^2. \quad (5.2)$$

Proof. We first show that

$$\text{cap}(A) = \inf\{\|f\|_{L^2}^2 : R_{1/2,1}f \geq 1 \text{ } \mu\text{-a.e. on } A, f \geq 0\}. \quad (5.3)$$

Afterwards, the claim (5.2) will follow from [25, Theorem 2.6.12]. Note that [25, Theorem 2.6.12] is formulated for Euclidean Bessel potentials, but its proof is purely variational and can thus be adapted to our setting. See also the analogous adaptation in [7, Lemma 5.5]. In order to show (5.3), note that the spectral theorem yields

$$R_{1/2,1}f = \sum_{k=1}^{\infty} \frac{1}{(1 + \lambda_k)^{1/2}} \langle f, \varphi_k \rangle \varphi_k \quad (5.4)$$

for any $f \in L^2(K)$, and also $R_{1/2,1}f \in \mathcal{F}$ since

$$\mathcal{E}_1(R_{1/2,1}f, R_{1/2,1}f) = \sum_{k=1}^{\infty} \frac{1 + \lambda_k}{1 + \lambda_k} \langle f, \varphi_k \rangle^2 = \|f\|_{L^2}^2.$$

Thus, for any $f \in L^2(K)$ with $R_{1/2,1}f \geq 1$ on A , the function $u = R_{1/2,1}f$ belongs to the family over which the capacity (5.1) minimizes, so that

$$\text{cap}(A) \leq \inf\{\|f\|_{L^2}^2 : R_{1/2,1}f \geq 1 \text{ } \mu\text{-a.e. on } A, f \geq 0\}.$$

This shows one inequality in (5.3). To prove the converse, for $u \in \mathcal{F}$ such that $u \geq 1$ μ -a.e. on A , set $f := g_+$, where $g := (I - \Delta)^{1/2}u$. Then, $f \in L^2(K)$ and $f \geq 0$. In addition, since $f \geq g$ and the kernel $r_{1/2,1}$ is nonnegative, it follows that

$$R_{1/2,1}f \geq R_{1/2,1}g = u,$$

and from (5.4)

$$\mathcal{E}_1(u, u) = \|g\|_{L^2}^2 \geq \|f\|_{L^2}^2.$$

Thus, any u in the infimum (5.1) yields a function f that belongs to the family on the RHS of (5.3), whence

$$\text{cap}(A) \geq \inf\{\|f\|_{L^2}^2 : R_{1/2,1}f \geq 1 \text{ } \mu\text{-a.e. on } A, f \geq 0\},$$

proving the equality (5.3). $\square$

With the help of the previous lemma we now obtain the smoothness of the measure ν^γ .

Lemma 17. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then, $\mathbb{P}$ -a.s. the LQG measure ν^γ charges no $\mathcal{E}_1$ -polar sets.*

Proof. First, we show that

$$\|R_{1/2,1}(\nu^\gamma)\|_{L^2}^2 < \infty. \quad (5.5)$$

Since $r_{1/2,1} \leq r_{1/2,0}$, the convolution property of these kernels, see [2, Corollary 2.11], implies

$$\begin{aligned} \|R_{1/2,1}(\nu^\gamma)\|_{L^2}^2 &= \int_K \int_K \int_K r_{1/2,1}(x, y) r_{1/2,1}(x, z) d\nu^\gamma(y) d\nu^\gamma(z) d\mu(x) \\ &\leq \int_K \int_K \int_K r_{1/2,0}(x, y) r_{1/2,0}(x, z) d\nu^\gamma(y) d\nu^\gamma(z) d\mu(x) \\ &= \int_K \int_K \int_K r_{1,0}(y, z) d\nu^\gamma(y) d\nu^\gamma(z). \end{aligned}$$

Further, note that $r_{1,0}(x, y)$ is uniformly bounded in $x, y \in K$ by virtue of [2, Proposition 2.8] and consequently the last integral above is indeed $\mathbb{P}$ -a.s. finite since $\nu^\gamma(K) < \infty$ $\mathbb{P}$ -a.s. by Corollary 11.

Next, let $\varepsilon > 0$ and let $E \subset K$ be a $\mathcal{E}_1$ -polar set. By definition, there exists an open set $A \subset K$ such that $E \subset A$ and $\text{cap}(A) < \varepsilon$. By applying the definition of $b(A)$ to $\tilde{\nu} := \|R_{1/2,1}(\mathbf{1}_A \nu^\gamma)\|_{L^2}^{-1} \mathbf{1}_A \nu^\gamma$ we obtain

$$\nu^\gamma(E) \leq \nu^\gamma(A) \leq b(A) \|R_{1/2,1}(\mathbf{1}_A \nu^\gamma)\|_{L^2} \quad \mathbb{P}\text{-a.s.}$$

Further, by Lemma 16 and since $R_{1/2,1}(\mathbf{1}_A \nu^\gamma) \leq R_{1/2,1}(\nu^\gamma)$

$$\nu^\gamma(E) \leq \text{cap}(A)^{1/2} \|R_{1/2,1}(\nu^\gamma)\|_{L^2} < \|R_{1/2,1}(\nu^\gamma)\|_{L^2} \varepsilon^{1/2}.$$

Now, the claim follows from (5.5) because $\varepsilon > 0$ was arbitrary chosen. $\square$

Corollary 18. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then, the LQG measure ν^γ on K is $\mathbb{P}$ -a.s. smooth.*

5.2. LBM on the Sierpinski gasket. The one-to-one correspondence between smooth measures and PCAFs, c.f. [12, Theorem 5.1.4], guarantees the existence up to equivalence of PCAFs of the time changed process $\{B_t^\gamma\}_{t \geq 0}$ on K given by

$$B_t^\gamma := B_{\tau_t^\gamma}, \quad \text{where } \tau_t^\gamma := \inf\{s > 0 : A_s^\gamma > s\}. \quad (5.6)$$

The main result of this section collects relevant features of this process, which we shall call the *Liouville Brownian Motion* (LBM) on K . In terms of notation, $(\mathcal{E}, \mathcal{F}_e)$ below refers to the the *extended Dirichlet space* of the Dirichlet form $(\mathcal{E}, \mathcal{F})$, c.f. [12, p.41].

Theorem 19. *Let $|\gamma| < \sqrt{d_H/C_0}$. Then, $\mathbb{P}$ -a.s. it holds that:*

- (i) *There exists a unique PCAF $\{A_t^\gamma\}_{t \geq 0}$ with Revuz measure ν^γ .*
- (ii) *The time-changed process $\{B_t^\gamma\}_{t \geq 0}$ given by (5.6) is a ν^γ -symmetric Hunt process with associated Dirichlet form*

$$\mathcal{E}^\gamma(u, v) = \mathcal{E}(u, v), \quad \mathcal{F}^\gamma = \mathcal{F}_e \cap L^2(K, \nu^\gamma). \quad (5.7)$$

Proof of Theorem 19. (i) The existence of the PCAF $\{A_t^\gamma\}_{t \geq 1}$ follows from [12, Theorem 5.1.4] and the smoothness of ν^γ from Corollary 18.

(ii) By virtue of [12, Theorem 6.2.1], the time-changed process (5.6) is a (symmetric) Hunt process and the associated Dirichlet form takes the expression (5.7) by virtue of [12, (6.2.22)] since ν^γ has full support $\mathbb{P}$ -a.s. by Corollary 18. $\square$

Finally, we mention that, if additionally ρ has bounded density with respect to μ , an adaptation of the arguments in the proof of [7, Theorem 5.7] shall lead to the approximation

$$A_t^\gamma = \lim_{\ell \rightarrow \infty} \int_0^t e^{\gamma h_\ell(B_s) - \frac{\gamma^2}{2} k_{d_S, \ell}(B_s, B_s)} ds$$

$\mathbb{Q}_\rho \otimes \mathbb{P}$ -a.s. for all $t \geq 0$. In particular, the PCAFs associated to ν_ℓ^γ will converge to the PCAF associated to ν^γ .

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